The Impact of Visual and Nonvisual Factors on Quality of Life and Adaptation in Adults with Visual Impairment

Ana Hernandez Trillo and Christine M. Dickinson

Purpose. Quality of life (QoL) questionnaires have been suggested as the most appropriate way to measure the effectiveness of low vision rehabilitation. This study investigated the relative contribution of visual and psychosocial factors to different aspects of QoL in people with low vision.

Methods. A total of 448 consecutive patients between the ages of 18 and 96 years, with best-corrected binocular visual acuity ≤6/18 and attending a low vision clinic, were recruited. Telephone delivery of previously validated questionnaires was used. The Low Vision Quality of Life (LVQOL), the Adaptation to Age-Related Vision Loss (AVL)-12, and the Keele Participation Restriction Questionnaire (KAP) questionnaires were considered as outcome measures for functional vision, adaptation to vision loss, and participation restriction, respectively. Personality (BFI-10), religious beliefs (SBI-15), social support (MOS), the mental and physical components of general health (the MCS and PCS of SF-12), well-being (WHO-5), use of magnifiers (MLVQ), understanding of their eye condition and satisfaction with the eye clinic (MLVQ), level of education, and financial status were all considered as predictive of QoL.

Results. Regression analysis found the PCS and MCS from SF-12 to be major predictors of LVQOL, AVL-12, and KAP scores. Although distance visual acuity and contrast sensitivity were predictors of LVQOL scores, "use of magnifiers" did not contribute to any of the QoL measures.

Conclusions. Nonvisual factors, such as physical and mental health, were found to be stronger predictors of QoL in people with low vision than visual factors such as contrast sensitivity and visual acuity, or the use of magnifiers. Researchers need to be aware when measuring QoL in a population with low vision that even vision-related QoL is strongly influenced by nonvisual variables. (*Invest Ophthalmol Vis Sci.* 2012;53:4234–4241) DOI:10.1167/iovs.12-9580

Low vision (LV) is defined as reduced visual acuity even when the individual is using the best possible optical correction[1] and is the consequence of untreatable ocular disease. In 2001, the World Health Organization (WHO)[2] published the International Classification of Functioning, Disability and Health, in which the terminology used to describe the consequences of disease was defined. Impairment describes the abnormalities in the structure or functioning of the body, whilst activity limitation refers to the difficulty in performing specific tasks in everyday life. Participation restriction refers to individuals' perception of the barriers experienced in life situations in relation to those impairments and activity limitations, in the context of their own personality, their adaptation to the visual loss, and the circumstances in which they live. Stucki et al.[3] explained this term as the complex interaction between the health condition, the individual, and the environment. It relates to the global consequences of impairment and/or activity limitation for the individual in terms of all aspects of life, that is, mobility, exchange of information, social relationships, education, work, leisure and spirituality, economic life, and civic and community life.[4]

Individuals' reaction to rehabilitation and their participation restriction are likely to be influenced by their adaptation to, and acceptance of, their visual loss. Negative reactions to the loss of vision include shock, anger, denial or disbelief, depression, self-pity, loss of self-esteem, and withdrawal,[5] but eventually many individuals reach a stage of realistic acceptance. Dodds et al.[6] also proposed that successful adjustment and adaptation to the visual impairment were related to appropriate rehabilitation, arguing that depression and anxiety seen in patients with LV were due to the difficulty in performing certain tasks that were previously taken for granted (loss of self-efficacy). Psychological adjustment to low vision can therefore have a considerable impact on individuals' quality of life (QoL).

The WHO[7] defines QoL as the "individual's perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns." QoL is used to define the physical, psychological, functional, social, and economic well-being of an individual: the impact of a disease may reduce health-related quality of life[8] or in this specific case, vision-related QoL. Quality of life is usually measured with questionnaires, and activity limitation and participation restriction are used as surrogate measures of QoL in many studies of health conditions: those relating to vision loss are no exception. These questionnaires have been used in recent years to assess the effectiveness of rehabilitation (such as the use of magnifiers and other low vision aids),[9–11] but they have not yet been successfully used to discriminate between different types of rehabilitation.[12–14]

The hypothesis of this study was that these outcome measures are potentially influenced by a wide range of (largely) uncontrollable nonvisual and psychosocial factors, which might mask the more limited changes produced by rehabilitation or other interventions.

Three questionnaires were treated as outcome measures representing the different aspects of QoL identified above: the Low Vision Quality of Life (the 25-item LVQOL), the Adaptation to Age-Related Vision Loss (the 12-item AVL), and the Keele

From the Faculty of Life Sciences, The University of Manchester, Manchester, United Kingdom.

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Corresponding author: Ana Hernandez Trillo, Faculty of Life Sciences, The University of Manchester, Carys Bannister Building, Dover Street, Manchester, M13 9PL, United Kingdom; ana.hernandeztrillo@manchester.ac.uk.

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Participation Restriction Questionnaire (KAP). The LVQOL questionnaire is divided into four subscales, three of which are concerned with functional vision ([1] distance vision, mobility, and lighting; [2] adjustment to the vision loss; [3] reading and fine work; and [4] other activities of daily living). Although the authors suggest a total score for this questionnaire,[15] each scale can be scored separately if researchers are interested in certain aspects of quality of life (e.g., adjustment).[16]

These instruments were chosen as representative of those which have been used to measure the effectiveness of rehabilitation, and indeed the LVQOL (Müskens R, et al. *IOVS* 2005;46:ARVO E-Abstract 297), AVL,[17] and KAP[18] have all been used in this way previously. However, to the authors' knowledge, the KAP questionnaire has not been used previously in a population with low vision.

A review of the literature was performed to identify the psychosocial factors that have been found to be related to chronic illnesses and low vision. Four specific databases (Medline, Embase, Cinahl Plus, PsycInfo) and a general one (Scopus) were used for the search, along with the Cochrane Library of Systematic Reviews. The search term *psychosocial AND health AND impairment OR* "*activity limitation*" *OR* "*participation restriction*" was used, and possible factors were extracted from the articles identified. These factors were then substituted in the search term in place of "psychosocial," which identified further articles. This process continued until no new factors were appearing.

The factors of age, sex, personality,[19] religious beliefs,[20] coping strategies,[21] social/family support,[22] level of education, financial status,[23] home circumstances (living alone),[24] the patients' understanding of their condition,[25] and depression/well-being[26] were identified as potential contributors to QoL. Other factors thought to be potentially influencing QoL were general health (mental and physical) and the visual status (eye disease, visual acuity [VA], and contrast sensitivity [CS]). Most of these factors have been investigated in studies of QoL in chronic illnesses[27,28] but not all of them have been considered previously when addressing QoL in people with low vision.

In the absence of strong support in previous work for one particular factor determining QoL in visual impairment, a comprehensive approach was taken in the current study, with the aim of determining the relative contribution of each of these factors, rather than concentrating on an in-depth study of just one of them. Any dominant factor identified as a result of this study could then be the subject of further, more detailed investigation. The aim of this study was therefore to assess as many factors as possible in a large unselected sample of low vision patients who attended a hospital low vision clinic delivering typical "low vision care," in order to identify those factors that predict QoL in people with low vision.

Methods

The aim was to survey all adult patients with visual impairment who attended the Manchester Royal Eye Hospital Low Vision Clinic for a first or follow-up appointment during a 1-year period (May 2009–August 2010). This period was to ensure that sufficient numbers of patients in low-incidence categories were included (e.g., young adults; congenital visual loss). The patient's unique hospital number was used to ensure that there was no duplicate recruitment. The patients were interviewed 1 month after their low vision assessment so that their low vision aids were up to date and their visual status was unlikely to have changed from that recorded at their last appointment. Therefore, recruitment took place during four 3-month periods, interspersed with 1 month "catch-up periods" to allow for the interviews of each group of patients to be completed, before the next cohort was recruited.

The optometrists at the low vision clinic approached eligible participants after they had received their usual care, at the end of the appointment. Individuals who met the following inclusion and exclusion criteria were recruited: aged 18 years or older, best-corrected binocular acuity <6/18 (0.48 logMAR) due to any eye condition, currently using an optical low vision aid (LVA) (electronic LVAs are not provided in the National Health Service, within the UK LV clinics), able to communicate fluently in English, having a phone, and with no obvious hearing difficulties in following an ordinary conversation. All those patients with learning disabilities or cognitive impairments of a degree that prevents independent living were also excluded (e.g., dementia, cerebral palsy). The age limit was set by the range over which questionnaires have been validated, and the language restriction was because some of the questionnaires were only available in English versions.

Participants were asked to undergo a telephone interview lasting approximately 1 hour: this avoided the need for the patient to return to the clinic but did require them to have adequate hearing. Informed written consent was obtained for all the subjects after explanation of the nature and possible consequences of the study, and the research followed the tenets of the Declaration of Helsinki.

The LVQOL, AVL-12, and KAP questionnaires were treated as the three outcome measures (dependent variables) used to evaluate the different aspects of QoL in people with LV. Further well-recognized questionnaires (Manchester Low Vision Quality of Life Questionnaire [MLVQ], Health Survey [SF-12], MOS Social Support Survey, Spiritual Beliefs Inventory [SBI-15R], Personality [BFI-5], Coping Strategies Questionnaire [Brief COPE], Depression Questionnaire [WHO-5]) were used to score the various psychosocial factors.

Various criteria were applied to select the different questionnaires to be used in the present study. These were practicality (i.e., quick, with straightforward easy-to-understand questions [since several questionnaires were to be administered]); validity (i.e., used and validated previously, ideally with heterogeneous populations); suitable for telephone delivery; phrased positively (e.g., WHO-5 well-being scale[29]); and shown to be feasible in an earlier pilot study.[30]

The scores from these questionnaires together with demographic data about sex, ethnicity, financial status, level of education, living alone or not, eye condition, age, and clinical measures, such as VA at distance and near, and CS, were treated as the independent variables (possible predictors of the outcomes) (Table 1).

All of the participants in this study were using at least one LVA. The interviews were conducted at least 4 weeks after recruitment, allowing time for the participants to become familiar with their most up-to-date LVA. The MLVQ[31] includes questions about the use of magnifiers (frequency, average time, and longest time of use of magnifiers) and this was used to measure visual rehabilitation; the other questions regarding knowledge of the eye condition (e.g., "name the part of the eye affected by your eye condition"), general knowledge about LV (e.g., "is it true that sitting too close to the TV causes your eyesight to worsen?"), and satisfaction with the LV service (i.e., "how satisfactory was the care you have received for your eye condition?") were designated as nonvisual factors.

Data Analysis

In a preliminary analysis of the data, nonparametric statistics (Spearman's ρ) were used to calculate the correlation between the different values obtained, because the data were ordinal. A value of $P < 0.05$ was considered significant, but this must be interpreted cautiously because of the multiple comparisons performed: no formal correction was made to significance level to reflect this point. Data were analyzed with SPSS 15.0 software (SPSS Inc., Chicago, IL). Several independent variables were statistically significantly correlated to each of the outcome measures; therefore, multiple linear regression analyses were performed with STATA 11.0 software (StataCorp, College Station, TX) by using the backwards-stepwise selection method. All the independent variables were entered into the regression equations. The use of

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Table 1. Summary of Outcome Measures and Independent Variables ("Contributing Factors") Investigated

Outcome Measures (Questionnaires)	Contributing Factors (Questionnaires)	Contributing Factors (Data Obtained from the Hospital Records/Patients)
LVQOL-25[15] (LV quality of life)	MLVQ[31] (use of magnifiers, knowledge of the eye condition, satisfaction with the LV service)	Best-corrected VA at distance with spectacles*
AVL-12[32] (adaptation to the vision loss)	MOS Social Support[33] (social support: tangible, affectionate, positive social interaction, emotional/informational)	VA at near with optimum magnifier*
KAP[34] (participation restriction)	SF-12[35] (physical and mental health)	Log CS*
	BFI-10[36] (personality: neuroticism, conscientiousness, agreeableness, extraversion, openness)	Age*
	SBI-15[37] (religious/spiritual beliefs)	Eye condition*
	Brief COPE[38] (coping strategies)	Sex*
	WHO-5[29] (well-being/depression)	Ethnicity*
		Postcode (financial status)*
		Living alone†
		Level of education†

* Data collected from hospital records.
† Data reported by the patient.

multiple regression analysis was also useful to tackle the problem of multiple testing and collinearity between measures.[39] Although the regression analysis does not totally avoid the drawback of multiple testing (i.e., contributing factors not being truly independent), it helps with collinearity by excluding from the analysis those variables that are linearly related (i.e., highly collinear).[40]

The rationale for the analysis was based on the approach followed by previous studies that have focused on quality of life in people with low vision[41–43]: to analyze the relationship between several factors and determine which ones were the main predictors of different outcome measures.

Some authors argue that classical testing theory (CTT) is not the best way to analyze questionnaire data.[44,45] They suggest the use of item response theory (IRT) as the alternative, and, in particular, the use of Rasch analysis. While CTT assigns ordinal numbers to items and gives a total score for the questionnaire, which is the sum across questions, IRT considers the differences in people's ability and items' difficulty; items are assigned a score according to these assumptions.[46] Rasch analysis is a probabilistic logistic model that creates logit values outlining each item difficulty and person ability.[46] In other words, Rasch analysis does not assume that all items have the same level of difficulty, but weights them accordingly. It is argued that this produces a more reliable and valid measurement, and a small number of low vision QoL instruments have been analyzed in this way.[44,47] Therefore, to ensure that the results in this study were consistent with this theory, each of the questionnaires was rescored with Rasch analysis and the new scores were used in the regression equation to see whether similar results would be obtained. Seven questionnaires were rescored with Rasch analysis: the LVQOL; the AVL-12; the KAP; the MOS Social Support (leaving out the first two questions about the number of friends and family and analyzing each category of social support [tangible, educational/informational, positive social interaction, affectionate] separately); the WHO-5 well-being questionnaire; the Brief COPE; and the SBI-15. The rest of the questionnaires could not be analyzed with this technique for various reasons: the SF-12 Health Survey had a different way of scoring each answer and this did not fit into the Rasch model; the MLVQ is subdivided into questions of different types (frequency of the use of magnifiers, average duration use of magnifiers, satisfaction with the LV service, knowledge about the eye condition, knowledge about LV in general), and therefore it was

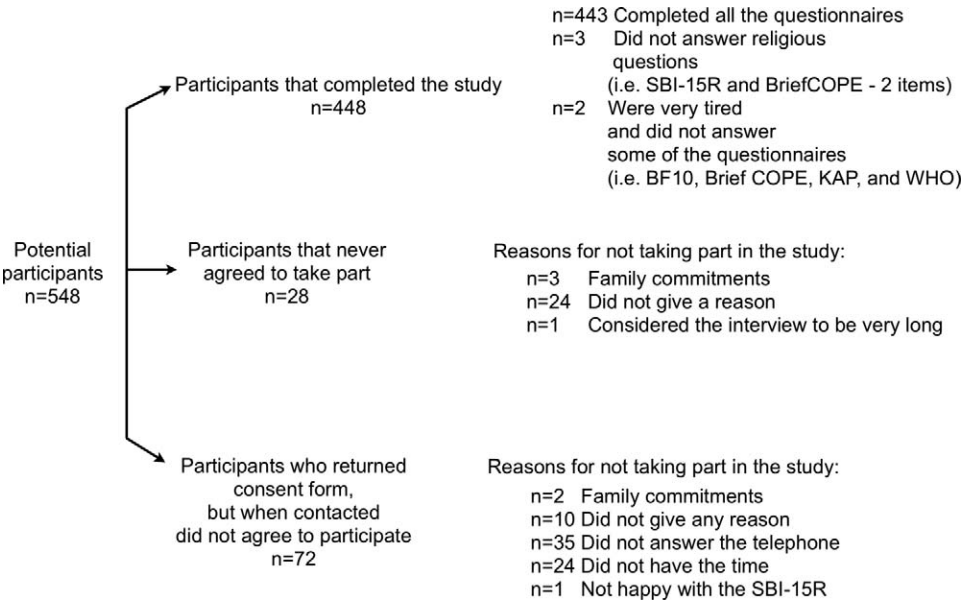


Figure. Recruitment of patients to the study from the total potential number of adult patients seen in the MREH Low Vision Clinic.

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Table 2. Characteristics of the Study Participants (Discrepancy in Numbers due to Missing Data)

		%	*n*
Sex	Female	59.6	267
	Male	40.4	181
Ethnicity	Caucasian	94.2	422
	South Asian	4.2	19
	Afro-Caribbean	1.1	5
	Mixed race	0.2	1
Higher education	No	327	73
	Yes	120	26.8
Living alone	Yes	48.7	218
	No	50.9	228
Postcode	Wealthy	19.0	85
	Comfortable	29.0	130
	Managing	8.9	40
	Poor	40.4	181
Eye condition	AMD	53.3	239
	Glaucoma	4.5	20
	Diabetic retinopathy	9.2	41
	Acquired retinal disorders (e.g., cystoid macular edema)	9.4	42
	Acquired nonretinal disorders (e.g., corneal dystrophy)	6.7	30
	Congenital retinal disorders (e.g., Stargardt's disease)	6.5	29
	Congenital nonretinal disorders (e.g., albinism)	4.7	21
	Cortical problems (e.g., bilateral amblyopia)	2.7	12
	Systemic disorders (e.g., stroke)	2.9	13

not possible to attribute a whole Rasch score for this questionnaire; and the BFI-10 had two items per personality trait, consequently producing very unreliable Rasch estimates. If Rasch analysis was not possible, the original scoring was kept in the reanalysis.

Results

During the recruitment period, a total of 923 adult patients had an appointment to attend the LV clinic at the Manchester Royal Eye Hospital (MREH); 199 never attended or cancelled their appointment at the LV clinic (even when the patients were contacted by one of the authors [A.H.T.] to remind them about their appointment), and 176 did not meet the inclusion criteria. There were therefore 548 potential participants for this study; 28 did not agree to take part and 520 returned the consent form. A total of 72 of these consenting subjects did not participate eventually, leaving 448 subjects who completed the study: overall, the participation rate was high (81.75%) (Figure).

The age of the participants completing the study ranged between 18 and 96 years, with a mean ± SD of 71.46 ± 17.74 years. The characteristics of the population are shown in Table 2.

The proportion of females to males was approximately 3:2; most were Caucasian, and more than 50% of participants had age-related macular degeneration: congenital disorders were present in only 11.2% of the population. The postcode was used in order to determine social/financial status from the "Financial ACORN" database (CACI Ltd., London, UK). Each postcode, which comprises a group of around 15 households, is assigned to 1 of 48 "types" (e.g., "retired wealthy suburban investors, many shares"). For the purposes of this study, these types were reduced to four categories to reflect "wealthy," "comfortable," "managing," and "poor" (Table 3). Approximately 40% of participants were classed as "poor," and about 50% of participants lived alone.

Table 3. Descriptive Statistics for the Questionnaires (Outcome Measures and Contributing Factors) in the Current Study

Questionnaires (Range of Possible Scores)	Minimum	Maximum	Mean ± SD
LVQOL Total (25–125)	26.00	121.00	71.74 ± 19.33
AVL-12 (0–36)	7.00	36.00	21 ± 4.62
KAP (0–11)	0.00	11.00	2 ± 2.35
SF-12 PCS (0–100)	14.77	65.56	39.02 ± 11.63
SF-12 MCS (0–100)	9.43	68.44	45.40 ± 11.02
MOS Total (19–95)	19.00	95.00	69.48 ± 19.26
SBI-15 (0–45)	0.00	45.00	21.02 ± 10.46
Extraversion (2–10)	2.00	10.00	6.65 ± 1.77
Agreeableness (2–10)	3.00	10.00	7.40 ± 1.47
Conscientiousness (2–10)	3.00	10.00	7.83 ± 1.50
Neuroticism (2–10)	2.00	10.00	5.28 ± 1.92
Openness (2–10)	2.00	10.00	6.52 ± 1.80
Brief COPE (0–84)	0.00	82.00	52.87 ± 11.59
WHO (0–25)	0.00	25.00	13.76 ± 6.57

The LVQOL, AVL-12, SF-12, MOS Social Support, SBI-15R, BFI-10, Brief COPE, and WHO are scored such that the higher the score that the participants achieve in these questionnaires, the better the LV quality of life, adaptation to the vision loss, and mental or physical health; the more the social support perceived; the stronger the spiritual/religious beliefs; the more dominant the personality trait; the more that positive coping strategies are used; and the greater the well-being of the participants. However, the KAP questionnaire is reversed, resulting in a higher score for those people who feel more restricted. The MLVQ is mostly scored so the higher the number, the greater the use of the magnifiers, but the satisfaction item is reversed, so the higher the score, the less satisfied the participant is. VA at distance and at near are measured in logMAR, therefore a higher number indicates worse visual acuity. On the other hand, a higher number indicates better (log) contrast sensitivity.

Table 3 shows the descriptive statistics for the outcome measures (LVQOL Total, AVL-12, and KAP) and the contributing factors (SF-12, MOS Social Support, SBI-15, BFI-10, Brief COPE, WHO).

Table 4 shows the correlation between the overall scores of AVL-12, LVQOL, and KAP. Although these correlations are in the expected direction (the better quality of life is associated with better adaptation and with less participation restriction) and statistically significant, they are not very strong. This suggests that they are, as intended, measuring different aspects of the influence of low vision on the patient's life. The strongest correlation (although only slightly so) was between AVL-12 and the LVQOL "Adjustment" subscale ($r = 0.554$, $P < 0.01$), and this was expected since these are dealing with the same aspect of the visual loss.

Table 4. The Correlation between the Total Scores on the Three Outcome Measures Used in the Study (Spearman's ρ)

	LVQOL (Total)	KAP
AVL-12	+0.537 ($P < 0.01$)	−0.360 ($P < 0.01$)
LVQOL (Total)		−0.434 ($P < 0.01$)

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Table 5. Correlation (Spearman's ρ) between the Outcome Measures (LVQOL, AVL-12, and KAP) and the Clinical Measures

	LVQOL Total		AVL-12		KAP	
	r	*P*	*r*	*P*	*r*	*P*
Use of magnifiers (MLVQ)	0.038	>0.05	0.018	>0.05	−0.026	>0.05
CS	**0.343**	**<0.01**	0.047	>0.05	−0.056	>0.05
VAD	**−0.347**	**<0.01**	−0.063	>0.05	0.081	>0.05
VAN	**−0.265**	**<0.01**	−0.045	>0.05	0.089	>0.05

Use of magnifiers is represented by the total score for the MLVQ questions on frequency, average use, and longest use of magnifiers. Bolding indicates values that are statistically significant.

The expectation and purpose of low vision rehabilitation is that it can be used to influence quality of life for low vision patients. We therefore correlated MLVQ (using the sum of the scores for frequency, average duration, and longest duration of use) and the other clinical visual measures (visual acuity at distance [VAD], visual acuity at near with magnification [VAN], and CS) with the three outcome measures (LVQOL, AVL-12, and KAP).

There was a statistically significant correlation between LVQOL and the three visual measures CS, VAD, and VAN, with Table 5 showing that the better the visual performance, the better the LV quality of life. However, no statistically significant correlation was found between any measure of vision and AVL-12 or KAP scores (Table 4). None of the outcome measures correlated significantly with the use of magnifiers, although there was a statistically significant, although weak, correlation between the LVQOL reading subscale and the use of magnifiers ($r = 0.222$; $P < 0.01$).

A correlation analysis of the other independent variables that were considered as contributory to the quality of life, and the outcome measures, showed several statistically significant correlations. Therefore, a regression analysis was performed in order to investigate the main predictors of low vision quality of life (LVQOL), adaptation to the vision loss (AVL-12) and participation restriction (KAP) in patients with visual impairment (VI). Where appropriate, the different questionnaire subscales (e.g., conscientiousness, neuroticism, agreeableness, openness, and extraversion for the BFI-10; positive social interaction, educational/informational, affectionate, and tangible social support for the different components of MOS; and the individual questions of the MLVQ) were used instead of the total score.

The models obtained with this method are illustrated in Tables 6, 7, and 8. All the data represented here were standardized to allow easier comparison between the regression coefficients. The alternative coefficients and *P* values obtained with Rasch analysis are also shown in Tables 6, 7, and 8.

Tables 6, 7, and 8 are the key findings in this investigation. In Table 6 it can be observed that the main contributing factors to LVQOL (Total) are physical and mental health, VAD, and CS. All of these factors have a relationship with LVQOL, such that the better these factors are, the better is the low vision quality of life. Another important predictor of low vision quality of life is the patient's satisfaction with the LV clinic. The remaining factors, although contributing to the variance (adjusted r^2) are not so important in predicting QoL in LV. In Table 7, both physical and mental health are again shown to play a major role as predictors of AVL-12. Other important contributors to positive adaptation are educational/informational support (e.g., having someone who understands your problems/someone who gives you good advice), general knowledge about low vision, and a higher level of education. However, male sex appears to be a predictor of worse adaptation to the vision loss. The rest of the factors, although contributing to the variance, are less important predictors of adaptation to vision loss. Table 8 shows that both physical and mental health are, once more, the most important predictors of KAP score. This

Table 6. Regression Model for LVQOL Including All Predictor Variables

	Coefficient		*t*		*P*	
	LVQoL-25	Rasch Analysis	LVQoL-25	Rasch Analysis	LVQoL-25	Rasch Analysis
Physical health (SF-12 PCS)	0.353	0.320	8.51	7.51	<0.0001	<0.0001
Mental health (SF-12 MCS)	0.309	0.296	6.72	6.41	<0.0001	<0.0001
Conscientiousness (BFI-10)	−0.040	−0.038	−1.03	−0.97	0.304	0.334
Knowledge of the eye condition (MLVQ)	−0.083	−0.082	−2.04	−1.92	0.042	0.056
Educational/informational support (MOS Social Support)	0.025	0.015	0.58	0.37	0.560	0.714
Frequency of the use of magnifiers (MLVQ)	−0.104	−0.086	−2.34	−1.89	0.020	0.060
Average length of time of use of magnifiers (MLVQ)	0.088	0.079	2.02	1.78	0.044	0.076
Age	0.005		1.99		0.047	
VAD	−0.729	−0.781	−4.47	−4.72	<0.0001	<0.0001
CS	0.559	0.453	4.81	3.83	<0.0001	<0.0001
Satisfaction (MLVQ)	−0.139	−0.120	−3.19	−2.74	0.002	0.006
Neuroticism (BFI-10)	−0.080	−0.101	−1.78	−2.22	0.076	0.027
Knowledge of the part of the eye affected by the eye condition (MLVQ)		−0.041		−1.01		0.315
Systemic disease		−0.467		−2.19		0.030

All the independent variables were introduced in the regression analysis, but only those that appeared to be predictors of the outcome measure are shown in the table. A blank cell in the table indicates that the variable was not a predictor in that analysis method. $r^2 = 0.4945$; adjusted $r^2 = 0.4756$. Rasch analysis: $r^2 = 0.4716$; adjusted $r^2 = 0.4485$.

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Table 7. Regression Model for AVL-12 Using All Predictor Variables

	Coefficient		*t*		*P*	
	AVL-12	**Rasch Analysis**	**AVL-12**	**Rasch Analysis**	**AVL-12**	**Rasch Analysis**
Physical health (SF-12 PCS)	0.257	0.266	5.72	5.95	<0.0001	<0.0001
Mental health (SF-12 MCS)	0.356	0.360	7.03	7.13	<0.0001	<0.0001
Positive social interaction (MOS Social Support)	0.096		1.29		0.197	
Educational/informational (MOS Social Support)	−0.267	−0.189	−3.43	−2.72	0.001	0.007
Openness (BFI-10)	0.098	0.100	2.22	2.28	0.027	0.023
Extraversion (BFI-10)	0.080	0.091	1.73	1.99	0.085	0.047
General knowledge about LV (MLVQ)	0.128	0.134	2.84	2.97	0.005	0.003
Neuroticism (BFI-10)	−0.093	−0.107	−1.85	−2.11	0.066	0.035
Satisfaction (MLVQ)	−0.107	−0.108	−2.20	−2.23	0.029	0.027
Affectionate (MOS Social Support)	0.140	0.149	2.00	2.22	0.046	0.027
Higher level of education	0.285		2.78		0.006	
Managing means	−0.305	−0.327	−2.06	−2.18	0.040	0.030
Male (sex)	−0.304	−0.320	−3.37	−3.52	0.001	0.001
Asian ethnicity	−0.482	−0.454	−1.94	−1.81	0.054	0.070
Congenital retinal dystrophies	0.361	0.374	1.74	1.78	0.083	0.076

All the independent variables were introduced in the regression analysis, but only those that appeared to be predictors of the outcome measure are shown in the table. A blank cell in the table indicates that the variable was not a predictor in that analysis method. $r^2 = 0.4078$; adjusted $r^2 = 0.3800$. Rasch analysis: $r^2 = 0.4042$; adjusted $r^2 = 0.3782$.

time it is seen that the worse the physical and mental health, the more restricted the individual is. Age appears to be another important predictor of participation restriction, indicating that the younger the individual, the more restricted he or she will feel. "Poor" financial status and conscientiousness are predictors of less participation restriction. Again, the remaining factors, although contributing to the variance (adjusted r^2) are not as important in predicting participation restriction.

The results obtained with Rasch analysis were very similar to the results obtained in the original analysis, with (largely) the same predictors appearing, and only minor changes in some coefficients.

Discussion

Measures of patient-reported functioning are growing in importance; the level of performance in everyday activities and patient's satisfaction cannot be determined without studies in quality of life.[48] Many studies have tried to define QoL[8] and relate it to life satisfaction, well-being, or happiness.[49] *Quality of life* is a difficult term to define, as shown by the large number of articles in the literature trying to label and classify it. WHO[7] has defined QoL as a subjective perception of a person's situation in life. It has been described as a multidimensional concept, influenced by several factors.[50] These factors include mental and physical health; economic situation; education; and friends and family. WHO[7] terms these factors as domains and identifies physical domain, psychological domain, level of independence, social relationships, environment, spirituality, religion, and personal beliefs.

When studying the effect of rehabilitation in patients with visual impairment, it is crucial to understand if QoL is being improved and whether clinicians are addressing patients' difficulties and concerns. QoL instruments have therefore been considered as appropriate outcome measures in rehabilitation studies. However, several studies[12–14] have been unable to identify differences in outcomes between different types of rehabilitation strategies. This study has found that vision-related QoL, as determined by three very different outcome

Table 8. Regression Model for KAP Using All Predictor Variables

	Coefficient		*t*		*P*	
	KAP	**Rasch Analysis**	**KAP**	**Rasch Analysis**	**KAP**	**Rasch Analysis**
Physical health (SF-12 PCS)	−0.313	−0.320	−6.46	−6.64	<0.0001	<0.0001
Mental health (SF-12 MCS)	−0.231	−0.229	−4.35	−4.36	<0.0001	<0.0001
Positive social interaction (MOS Social Support)	−0.100	−0.224	−2.11	−3.81	0.035	<0.001
Depression (WHO)	0.133	−0.149	−2.23	−2.52	0.026	0.012
Extraversion (BFI-10)	0.096	0.115	2.15	2.59	0.032	0.010
Conscientiousness (BFI-10)	−0.111	−0.089	−2.62	−2.14	0.009	0.033
General knowledge about LV (MLVQ)	−0.082		−1.82		0.069	
VA at near	0.320	0.343	1.92	2.06	0.056	0.040
Age	−0.012	−0.011	−4.35	−4.24	<0.0001	<0.0001
Coping strategies (Brief Cope)	−0.123		−2.47		0.014	
Congenital retinal dystrophies	−0.416	−0.383	−2.01	−1.81	0.046	0.063
Poor means	−0.230	−0.383	−2.63	−1.87	0.009	0.063
Systemic condition	0.413		1.82		0.070	

All the independent variables were introduced in the regression analysis, but only those that appeared to be predictors of the outcome measure are shown in the table. A blank cell in the table indicates that the variable was not a predictor in that analysis method. $r^2 = 0.4023$; adjusted $r^2 = 0.3761$. Rasch analysis: $r^2 = 0.3987$; adjusted $r^2 = 0.3763$.

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Table 9. Data regarding PCS and MCS Scores from the SF-12 in the Current Study in Comparison to Previous Studies

Study	Population Type	*n*	Age (y)	PCS	MCS	
Ware et al.[51]	Reference general US population	408	65–74	43.65 ± 11.02	52.10 ± 9.53	
Lamoureux et al.[52]	VI with diabetic retinopathy	45	67.5 ± 13.1	34.4 ± 11.8	46.6 ± 12.3	
Cahill et al.[53]	AMD undergoing macular translocation	50	76.9 ± 5.7	44.8 ± 9.2	49.3 ± 10.4	Pretreatment
				44.2 ± 9.5	50.8 ± 10.3	1 year post treatment
Hassell et al.[54]	VI with AMD	106	83.6 ± 7.6	38.1 ± 11.5	48.6 ± 10.9	
Kuyk et al.[55]	Registered blind enrolled in comprehensive in-patient rehabilitation	206	70.2 ± 12.6	44.1 ± 10.8	49.2 ± 11.7	Pre rehabilitation
				40.3 ± 11.9	51.9 ± 10.9	Post rehabilitation
Current study	Mixed VI	448	71.47 ± 17.7	39.02 ± 11.6	45.40 ± 11.0	

measures, is strongly dependant on factors unrelated to the use of low vision aids, in particular the mental and physical components of general health.

The SF-12 is scored such that higher numbers indicate better physical and mental health. The physical component score (PCS) and mental component score (MCS) each range from 0 to 100, with a mean score of 50 (SD, 10) for a representative sample of the US general population.[51] The PCS and MCS of the population in the present study is shown in Table 9, along with results from comparable populations.

The data indicate that both PCS and MCS in this study are below the general population mean indicated by Ware et al. (2002), but it is worth comparing the results with those of other elderly visually impaired populations. The MCS score is lower than that of other comparable groups; PCS is better than that of a previously described diabetic group[52] or a group with AMD of much higher average age.[54] Evidence from a study treatment[53] suggested no change to either PCS or MCS; rehabilitation appeared to give a minimal improvement in MCS, although the reduction in PCS was unexplained.[55] This suggests that "self-management" programs may be beneficial for visually impaired individuals, not only to deal with the visual problem but also to educate them in how to take better care of themselves to improve their physical health. When older individuals can be persuaded to take part in such schemes, they have shown significant increases in PCS and MCS scores.[56] Other factors, such as knowledge of the eye condition and education, some traits of personality, and some forms of social support, are also predictors of QoL but to a lesser extent. The visual impairment itself, in the form of VAD and CS, is a strong and significant predictor of the LVQOL; however, use of magnifiers makes only a minor contribution to LVQOL and does not predict AVL or KAP at all. Optimum VA at near (with magnification) was a strong predictor of participation restriction.

Patient-centered outcome measures are important to understand whether patients' concerns and difficulties are being addressed, and they are increasingly used to measure the effectiveness of rehabilitation. The results of this study, however, suggest that researchers investigating the effectiveness of any intervention should take great care when choosing questionnaires to measure QoL for people with LV. A specific intervention influences a relatively small part of the patients' life, and individuals have other issues in addition to their poor sight. It would be ambitious to think that providing very specific low vision interventions, such as supplying magnifiers, will significantly improve all aspects of QoL. Researchers need to be aware that even vision-related QoL is strongly influenced by nonvisual factors, and in particular physical and mental health.

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References

1. Dickinson C. *What is low vision?* In: *Low Vision Principles and Practice*. Edinburgh: Butterworth and Heinemann; 1998:3.
2. World Health Organization. *International Classification of Functioning, Disability, and Health (ICF)*. Geneva, Switzerland: World Health Organization; 2001.
3. Stucki G, Cieza A, Ewert T, Kostanjsek N, Chatterji S, Ustun B. Application of the International Classification of Functioning, Disability and Health (ICF) in clinical practice. *Dis Rehabil*. 2002;24:281–282.
4. Kearney PM. The International Classification of Functioning, Disability and Health (ICF) and nursing. *J Adv Nurs*. 2003;46:162–170.
5. Allen MN. The meaning of visual impairment to visually impaired adults. *J Adv Nurs*. 1989;14:640–646.
6. Dodds AG, Bailey P, Pearson A, Yates L. Psychological factors in acquired visual impairment: the development of a scale of adjustment. *J Vis Impair Blind*. 1991;85:306–310.
7. World Health Organization. Quality of Life. The world health organization quality of life assessment (WHOQOL): position paper from the world health organization. *Soc Sci Med*. 1995;41:1403–1409.
8. Galloway S. Well-being and quality of life: measuring the benefits of culture and sport: a literature review and thinkpiece. *Scott Executive Soc Res*. 2005;4–97.
9. Stelmack J, Tang C, Reda DJ, Rinne S, Mancil RM, Massof RW. Outcomes of the Veterans Affairs Low Vision Intervention Trial (LOVIT). *Arch Ophthalmol*. 2008;126:608–617.
10. Pearce E, Crossland MD, Rubin GS. The efficacy of low vision device training in a hospital-based low vision clinic. *Br J Ophthalmol*. 2010;95:105–108.
11. Court H, Ryan B, Bunce C, Margrain TH. How effective is the new community-based Welsh low vision service? *Br J Ophthalmol*. 2011;95:178–184.
12. de Boer M, Twisk J, Moll A, Volker-Dieben H, de Vet H, van Res G. Outcomes of low-vision services using optometric and multidisciplinary approaches: a non-randomized comparison. *Ophthalmic Physiol Opt*. 2006;26:535–544.
13. La Grow SJ. The effectiveness of comprehensive low vision services for older persons with visual impairments in New Zealand. *J Vis Impair Blind*. 2004;98:679–692.
14. Reeves B, Harper R, Russell W. Enhanced low vision rehabilitation for people with age-related macular degeneration: a randomised controlled trial. *Br J Ophthalmol*. 2004;88:1443–1449.
15. Wolffsohn JS, Cochrane AL. Design of the low vision quality-of-life questionnaire (LVQOL) and measuring the outcome of low-vision rehabilitation. *Am J Ophthalmol*. 2000;130:793–802.

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16. de Boer MR, Terwee CB, de vet HCW, Moll AC, Volker-Dieben HJ, van Rens GHB. Evaluation of cross-sectional and longitudinal construct validity of two vision-related quality of life questionnaires: the LVQOL and VCM1. *Qual Life Res*. 2006;15:233–248.
17. Tolman J, Hill RD, Kleinschmidt JJ, Gregg CH. Psychosocial adaptation to visual impairment and its relationship to depressive affect in older adults with age-related macular degeneration. *Gerontologist*. 2005;45:747–753.
18. Wilkie R, Thomas E, Mottram S, Peat G, Croft PR. Onset and persistence of person-perceived participation restriction in older adults: a 3-year follow-up study in the general population. *Health Qual Life Outcomes*. 2008;6:92. doi:10.1186/1477-7525-6-92.
19. Chapman BP, Duberstein PR, Sorensen S, Lyness JM. Personality and perceived health in older adults: the five factor model in primary care. *J Gerontol B Psychol Sci Soc Sci*. 2006;11:362–365.
20. Helm HM, Hays JC, Flint EP, Koenig HG, Blazer DG. Does private religious activity prolong survival: a six-year follow-up study of 3,851 older adults. *J Gerontol A Biol Sci Med Sci*. 2000;55:400–405.
21. Carver CS, Pozo C, Harris SD, et al. How coping mediates the effect of optimism on distress: a study of women with early stage breast cancer. *J Pers Soc Psychol*. 1993;65:375–390.
22. Curtis R, Groarke A, Coughlan R, Gsel A. The influence of disease severity, perceived stress, social support and coping in patients with chronic illness: a 1 year follow up. *Psychol Health Med*. 2004;9:456–475.
23. Smith DM, Langa KM, Kabeto MU, Ubel PA. Health, wealth, and happiness: financial resources buffer subjective well-being after the onset of a disability. *Psychol Sci*. 2005;16:663–666.
24. Verstraten PFJ, Brinkmann WLJH, Stevens NL, Schouten JSAG. Loneliness, adaptation to vision impairment, social support and depression among visually impaired elderly. *Int Congress Ser*. 2005;1282:317–321.
25. Kupfer C. The Low Vision Education Program: improving quality of life. *Optom Vis Sci*. 1999;76:729–730.
26. Lopes AA, Bragg J, Young E, et al. Depression as a predictor of mortality and hospitalization among hemodialysis patients in the United States and Europe. *Kidney Int*. 2002;62:199–207.
27. Allen R, Rowe M. Spirituality as a means of coping with chronic illness. *Am J Health Stud*. 2004;19:62–67.
28. Hoth KF, Christensen AJ, Ehlers SL, Raichle KA, Lawton WJ. A longitudinal examination of social support, agreeableness and depressive symptoms in chronic kidney disease. *J Behav Med*. 2007;30:69–76.
29. World Health Organization. Well-being measures in primary health care: The DepCare Project. Stockholm, Sweden: World Health Organization Regional Office for Europe; 1998.
30. Hernandez-Trillo A, Dickinson CM. *The Effect of Psychosocial Factors on Quality of Life in Visually Impaired Patients. Department of Optometry*. Manchester: University of Manchester; 2007.
31. Harper R, Doorduyn K, Reeves B, Slater L. Evaluating the outcomes of low vision rehabilitation. *Ophthalmic Physiol Opt*. 1999;19:3–11.
32. Horowitz A, Reinhardt JP, Raykov T. Development and validation of a short-form adaptation of the age-related vision loss scale: the AVL12. *J Vis Impair Blind*. 2007;101:146–159.
33. Sherbourne C, Stewart A. The MOS Social Support Survey. *Soc Sci Med*. 1991;32:705–714.
34. Wilkie R, Peat G, Thomas E, Hooper H, Croft PR. The Keele Assessment of Participation: a new instrument to measure participation restriction in population studies. Combined qualitative and quantitative examination of its psychometric properties. *Qual Life Res*. 2005;14:1889–1899.
35. Gandek B, Ware JE, Aaronson NK, et al. Cross-validation of item selection and scoring for the SF-12 Health Survey in nine countries: results from the IQOLA Project. *J Clin Epidemiol*. 1998;51:1171–1178.
36. Rammstedt B, John O. Measuring personality in one minute or less: a 10-item short version of the big five inventory in English and German. *J Res Pers*. 2007;41:203–212.
37. Holland JC, Kash KM, Passik S, et al. A brief spiritual inventory for use in quality of life research in life-threatening illness. *Psychol-Oncol*. 1998;7:460–469.
38. Carver CS. You want to measure coping but your protocol's too long: consider the brief COPE. *Int J Behav Med*. 1997;4:91–100.
39. Gelman A, Hill J, Yajima M. Why we (usually) don't have to worry about multiple comparisons. *J Res Educ Eff*. 2009;1–30.
40. Chen X, Ender PB, Mitchell M, Wells C. Regression with stata. In: *Stata Web Books*. Los Angeles, CA: UCLA Academic Technology Services; 2003.
41. Lamoureux EL, Hassell JB, Keeffe JE. The determinants of participation in activities of daily living in people with impaired vision. *Am J Ophthalmol*. 2004;137:265–270.
42. Rubin GS, Bandeen-Roche K, Huang GH, et al. The association of multiple visual impairments with self-reported visual disability: SEE project. *Invest Ophthalmol Vis Sci*. 2001;42:64–72.
43. O'Conner PM, Lamourex EL, Keeffe JE. Predicting the need for low vision rehabilitation services. *Br J Ophthalmol*. 2008;92:252–255.
44. Pesudovs K. Patient-centred measurement in ophthalmology—a paradigm shift. *BMC Ophthalmol*. 2006;6:25. doi:10.1186/1471-2415-6-25.
45. Massof R, Rubin G. Visual function assessment questionnaires. *Surv Ophthalmol*. 2001;45:531–548.
46. Bond TG, Fox CM. *Applying the Rasch Model: Fundamental Measurement in the Human Sciences*. Mahwah, New Jersey: Lawrence Erlbaum Associates; 2001.
47. Ryan B, Court H, Margrain TH. Measuring low vision services outcomes: Rasch analysis of the seven-item National Eye Institute Visual Function Questionnaire. *Optom Vis Sci*. 2008;85:112–121.
48. Mangione CM, Gutierrez PR, Lowe G, Orav EJ, Seddon JM. Influence of age-related maculopathy on visual functioning and health-related quality of life. *Am J Ophthalmol*. 1999;128:45–53.
49. Farquhar M. Definitions of quality of life: a taxonomy. *J Adv Nurs*. 1995;22:502–508.
50. Camfield L, Skevington SM. On subjective well-being and quality of life. *J Health Psychol*. 2008;13:764–775.
51. Ware J, Kosinski M, Turner-Bowker D, Gandek B, eds. *User's Manual for the SF-12v2 Health Survey with a Supplement Documenting SF-12 Health Survey*. Lincoln, RI: QualityMetric Incorporated; 2002.
52. Lamoureux EL, Hassell JB, Keeffe JE. The impact of diabetic retinopathy on participation in daily living. *Arch Ophthalmol*. 2004;122:84–88.
53. Cahill MT, Stinnett SS, Banks AD, Freedman SF. Quality of life after macular translocation with 360 peripheral retinectomy for age-related macular degeneration. *Ophthalmol*. 2005;112:144–151.
54. Hassell JB, Lamoureux EL, Keeffe JL. Impact of eye related macular degeneration on quality of life. *Br J Ophthalmol*. 2006;90:593–596.
55. Kuyk T, Liu L, Elliott JL, Grubbs HE. Health-related quality of life following blind rehabilitation. *Qual Life Res*. 2008;17:497–507.
56. Frosch DL, Rincon D, Ochoa S, Mangione CM. Activating seniors to improve chronic disease care: results from a pilot intervention study. *J Am Geriatr Soc*. 2010;58:1496–1503.

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